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New chapter

" Polymer solution "

To make polymer solution, there are two stage we seen as like -

1. First stage ⇒ slow penetration of solvent molecule into the interstics of the polymer coil and forcing them swell. (Extend)

- pseudo - induction period.

2. Second stage



Solid polymer — Swollen polymer in solvent — Solvated polymer molecules diffusing out of

Q: polymer solution's viscosity is high - why?

Thermodynamics of polymer dissolution

Dissolution process is a change of state (ΔH enthalpy, entropy)

- $\Delta H \rightarrow$ minimum (−)
- $\Delta S \rightarrow$ maximum

Spontaneously solution process

During solution process -

(1) Enthalpy can increase (+ΔH), decrease (−ΔH) or remain unchange

(11) Entropy can decrease or increase

(111) (−ΔH) and (+ΔS) both support

changes of state.

(iv) $(+\Delta H)$ & $(-\Delta S)$ both parameter oppose change.

Enthalpy Entropy

$$\Delta F = \Delta H - T\Delta S$$

(Exothermic & Endothermic)

Effect of molecular weight on solubility:

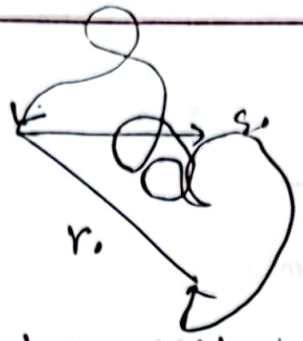
MW of polymer is not fixed rather a range and thereby we take an average value of MW.

gel-sol system: A system where equilibrium exists between solution phase and swollen solute phase is reffered as gel-sol system.

~~gel-sol sy~~

The solubility of gel-sol system can be increased by adding solvent (good) (binary)

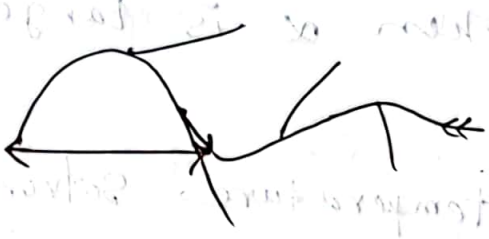
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a) Random coil

r_0 = End to end distance of a polymer chain

S_0 = Radius of gyration



$$(\overline{r_0})^{1/2} = \bar{l}\,(n)^{1/2}$$

$(\overline{r_0})^{1/2}$ = root-mean-square end to end distance

$\bar{l}$ = Average length of segment

n = no. of segments in chain

(I) Cohesive force

(II) Dispersion force

$$(\bar{r})^{1/2} = \alpha (\bar{r}_0)^{1/2}$$

α means factor that nature of solvent

When α is large, that is good solvent

At which temperature solvent is going to solution, that phase $\alpha = 1$.

⇒ Amorphous solid gets soluble precurs than crystalline solid (polymer)

⇒ The way of crystaline polymer to get soluble :-

(I) By heating crystalline polymer in its melting point

(II) By bringing condition so as to destroy the long range order

The free energy change for crystaline polymer-

$$\Delta G = (\Delta H_m + \Delta H_f) - T(\Delta S_m + \Delta S_f)$$

Where, m = mixing

f = fusion

Solubility parameter

Heat of mixing of two liquid

$$\Delta H_m = V_m \left[\left(\frac{\Delta E_L}{V_L} \right)^{1/2} - \left(\frac{\Delta E_1}{V_1} \right)^{1/2} \right]^2 Q_1 Q_2$$

$$= V_m [\delta_2 - \delta_1]^2 Q_1 Q_2$$

ΔH_m = Heat of mixing

V_m = total volume

ΔE_1, ΔE_2 = Energy of vapourization

$V_1 V_2$ = molar volume of two liquid

$Q_1 Q_2$ = volume fraction of two liquid

$$\left(\frac{\Delta E}{V} \right)^{1/2} = \delta$$

$$\Rightarrow \frac{\Delta H_m}{V_m Q_1 Q_2} = (\delta_2 - \delta_1)$$

When; $\delta_1 = \delta_2$ then

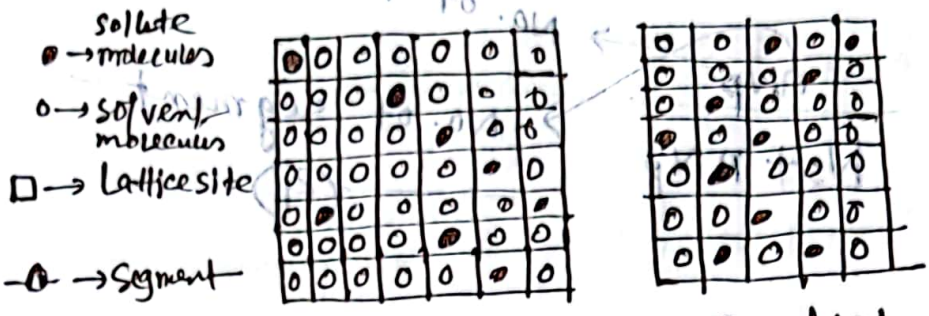
$$\frac{\Delta H_m}{V_m Q_1 Q_2} = 0$$

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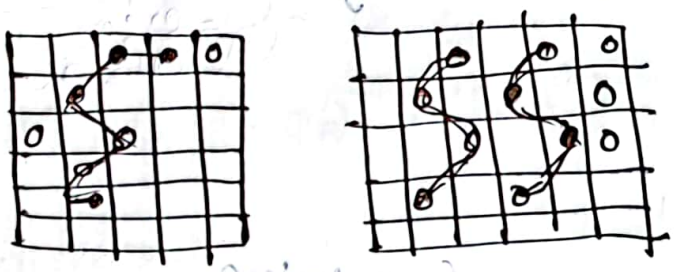
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⇒ Flory Huggins theory of polymer solution :



for low concentration

for high concentration



⇒ polymer

Fig: Two dimensional quasi-crystalline model and polymer

$$\Delta S_m = k \ln W \quad \text{——— (1)}$$

$$\Delta S_m = -k (N_s \ln \phi_s + N_p \ln \phi_p) \quad \text{——— (2)}$$

where, s = solvent ; p = polymer,

ϕ_s = volume fraction for solvent | ϕ_p = volume fraction for polymer

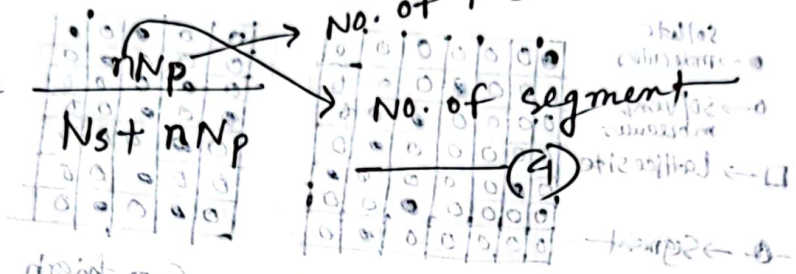
$$\phi_{s1} = \frac{N_s}{(N_s + n N_p)} \quad \text{(3)}$$

N_s → no. of solvent molecule

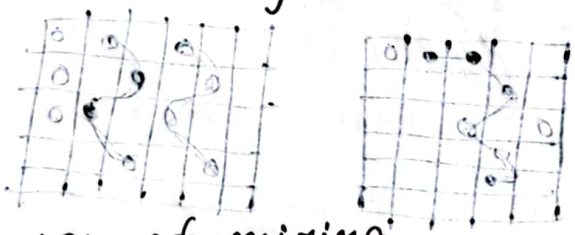
$$\phi_p = \frac{nN_p}{N_s + nN_p}$$

nN_p → No. of polymer molecule

$N_s + nN_p$ → No. of segment (4)



Large molecule has lower movement and thereby has less entropy



Enthalpy of mixing,

$$\Delta H_m = kT\chi N_s \phi_p \quad \text{(5)}$$

χ (ky) → solvent parameter

χ = low ⇒ good solvent

χ = high ⇒ poor solvent

$\bar{n}$ → average

$$\Delta F_m = KT(\chi N_s \phi_p + N_s \ln\phi_s + N_p \ln \phi_p) \quad \text{—(6)}$$

$$\Delta F = \Delta F_{conformation} + \Delta F_{contact} \quad \text{—(7)}$$

Where,

$$\Delta F_{conformation} = RT\left[\ln(1-\phi_p) + \left(1-\frac{1}{n}\right)\phi_p\right]$$

$$\Delta F_{contact} = RT[\chi \phi_p^2]$$

$$\therefore \Delta F_m = RT\left[\ln(1-\phi_p) + \left(1-\frac{1}{n}\right)\phi_p + \chi\phi_p^2\right] \quad \text{—— 10}$$

↳ for monodispersed

monodispersed ⇒ same MW of monomer in polymer segment

polydispersed ⇒ different

$\bar{n}$ → average

$$\Delta F_m = RT\left[\ln(1-\phi_p) + \left(1-\frac{1}{\bar{n}}\right)\phi_p + \chi\phi_p^2\right] \quad \text{—(12)}$$

↳ for polydispersed

critical value

(2) $(\chi)_c = \frac{1}{2}n + \frac{1}{\sqrt{n}} + \frac{1}{2}$

(1) $(\phi_p)_c \geq \frac{1}{1+\sqrt{n}}$

↳ Critical composition

$$\Delta F_{contact} = RT\left[\chi\phi_p^2 + \chi'\phi_p^3 + \chi''\phi_p^4 + \cdots\right]$$

(Modified equation)

or

$$\pi \bar{V}_s = -\Delta F_m$$

$\bar{V}_s$ → partial molar volume of solvent

$$\therefore -\pi\bar{V}_s = RT\left[\ln(1-\phi_p) + \left(1-\frac{1}{n}\right)\phi_p + \chi\phi_p^2\right]$$

(11) $\Rightarrow \frac{\pi}{V_s} = \frac{-RT}{V_s}\left[\ln(1-\phi_p) + \left(1-\frac{1}{n}\right)\phi_p + \chi\phi_p^2\right]$

~~For~~ In dilute solution, $\phi_p \ll 1$

and $\ln(1-\phi_p) \approx -\left[\phi_p + \frac{\phi_p^2}{2} + \frac{\phi_p^3}{3} + \cdots\right]$

$$\pi = \frac{RT}{V_s}\left[\frac{\phi_p}{n} + \left(\frac{1}{2} - \chi\right)\phi_p^2 + \frac{1}{3}\phi_p^3 + \cdots\right]$$

$$\Rightarrow \pi = RT\left[\frac{\phi_p}{n\bar{V}_s} + \left(\frac{1}{2} - \chi\right)\frac{\phi_p^2}{\bar{V}_s} + \frac{1}{3}\frac{\phi_p^3}{\bar{V}_s} + \cdots\right]$$

$$\pi = RT\left[\frac{c}{M_n} + \frac{\bar{V}_p}{\bar{V}_s}\left(\frac{1}{2} - \chi\right)c^2 + \cdots\right]$$

$$\pi = \left[\left(\frac{RT}{M_n}\right)c + RT\left(\frac{\bar{V}_p^2}{\bar{V}_s}\right)\left(\frac{1}{2} - \chi\right)c^2 + \cdots\right]$$

$$\frac{\pi}{c} = RT\left(\frac{1}{M_n} + Bc\right)$$

$$\left(\frac{\pi}{c}\right)_{c \to 0} = \frac{RT}{M_n}$$